SEAT BELT WARNING ALARM

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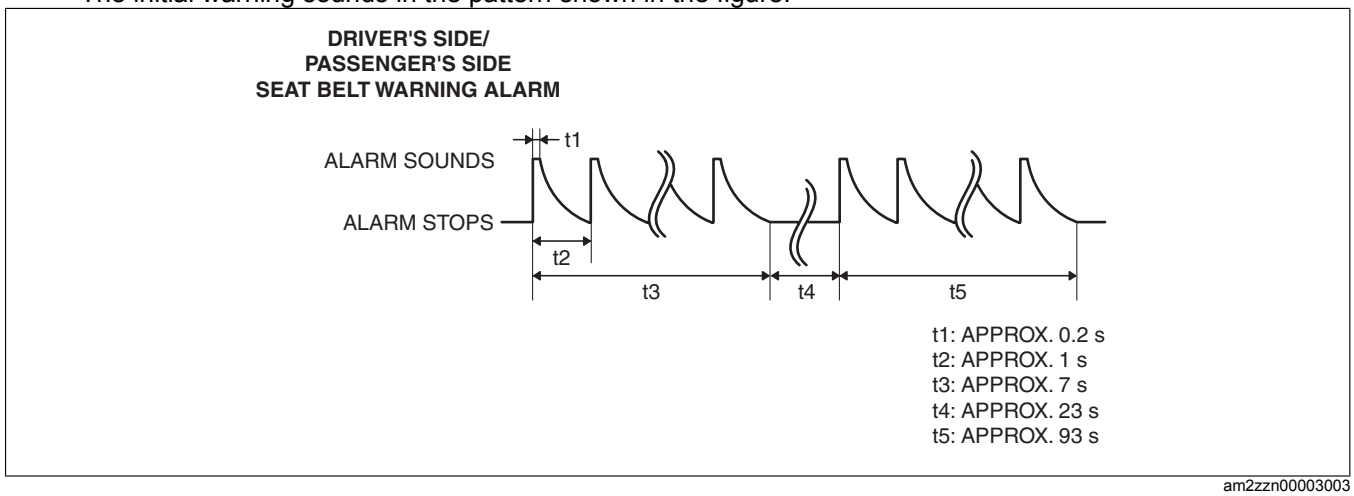
Purpose
- The seat belt warning alarm warns the driver that a seat belt is not fastened.

Function
- The instrument cluster controls the seat belt warning alarm based on the following signals.

Signal name	Sending module/part name	Communication method
Vehicle speed signal	PCM	HS-CAN
Driver's side/passenger's side buckle switch signal	SAS control module	
Occupancy sensor signal		
Rear seat buckle switch signal	BCM	

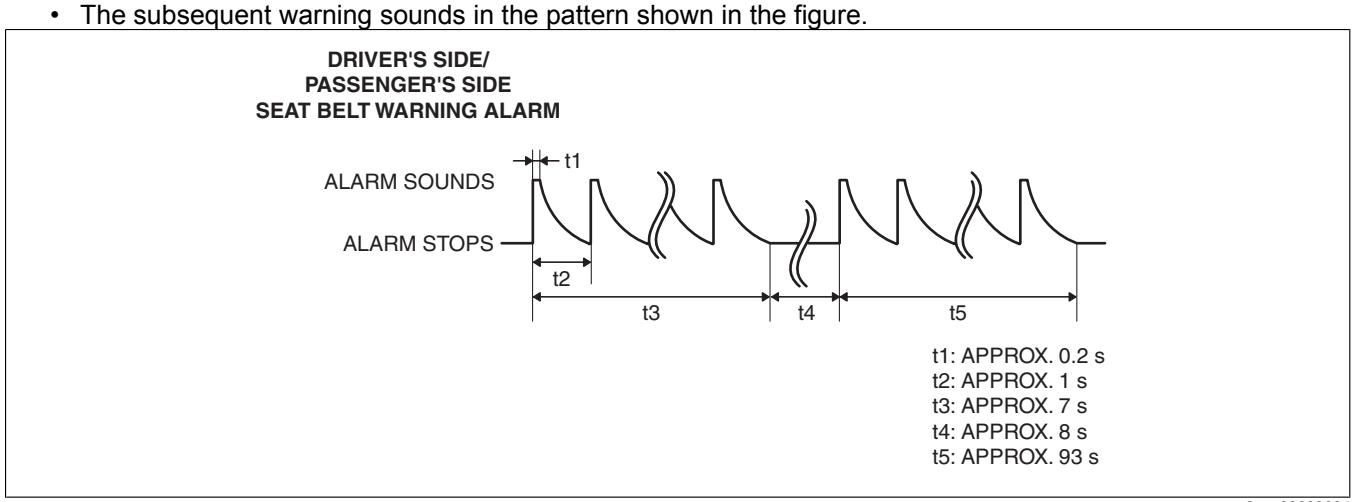
Driver's side/passenger's side seat belt warning alarm sound function
- The driver's side/passenger's side seat belt warning alarm has an initial warning and subsequent warning.

Initial warning
- When all of the following conditions are met, the alarm sounds.
  - — Vehicle speed is **approx. 20 km/h {12 mph} or more**
  - — Driver's side/passenger's side buckle switch on signal (seat belt unfastened) is detected
  - — Occupancy sensor on signal (occupant in passenger seat) is detected (passenger's side seat belt warning alarm only)
- The initial warning sounds in the pattern shown in the figure.



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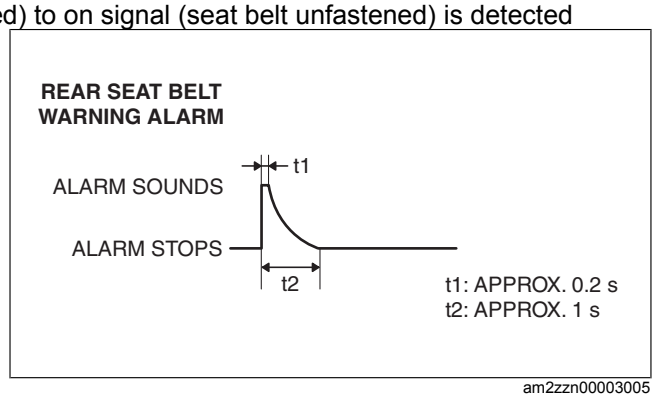
Subsequent warning
- When all of the following conditions are met, the alarm sounds.
  - — Vehicle speed is **approx. 20 km/h {12 mph} or more**
  - — Occupancy sensor on signal (occupant in passenger seat) is detected (passenger seat belt warning alarm only)
  - — After initial warning operates, buckle switch off signal (seat belt fastened) to on signal (seat belt unfastened) is detected
- The subsequent warning sounds in the pattern shown in the figure.



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Rear seat belt warning alarm sound function (with rear seat belt warning alarm)

- When all of the following conditions are met, the alarm sounds.
  - — Ignition switch ON (engine off or on)
  - — Rear seat buckle switch off signal (seat belt fastened) to on signal (seat belt unfastened) is detected
- The alarm sounds in the pattern shown in the figure.



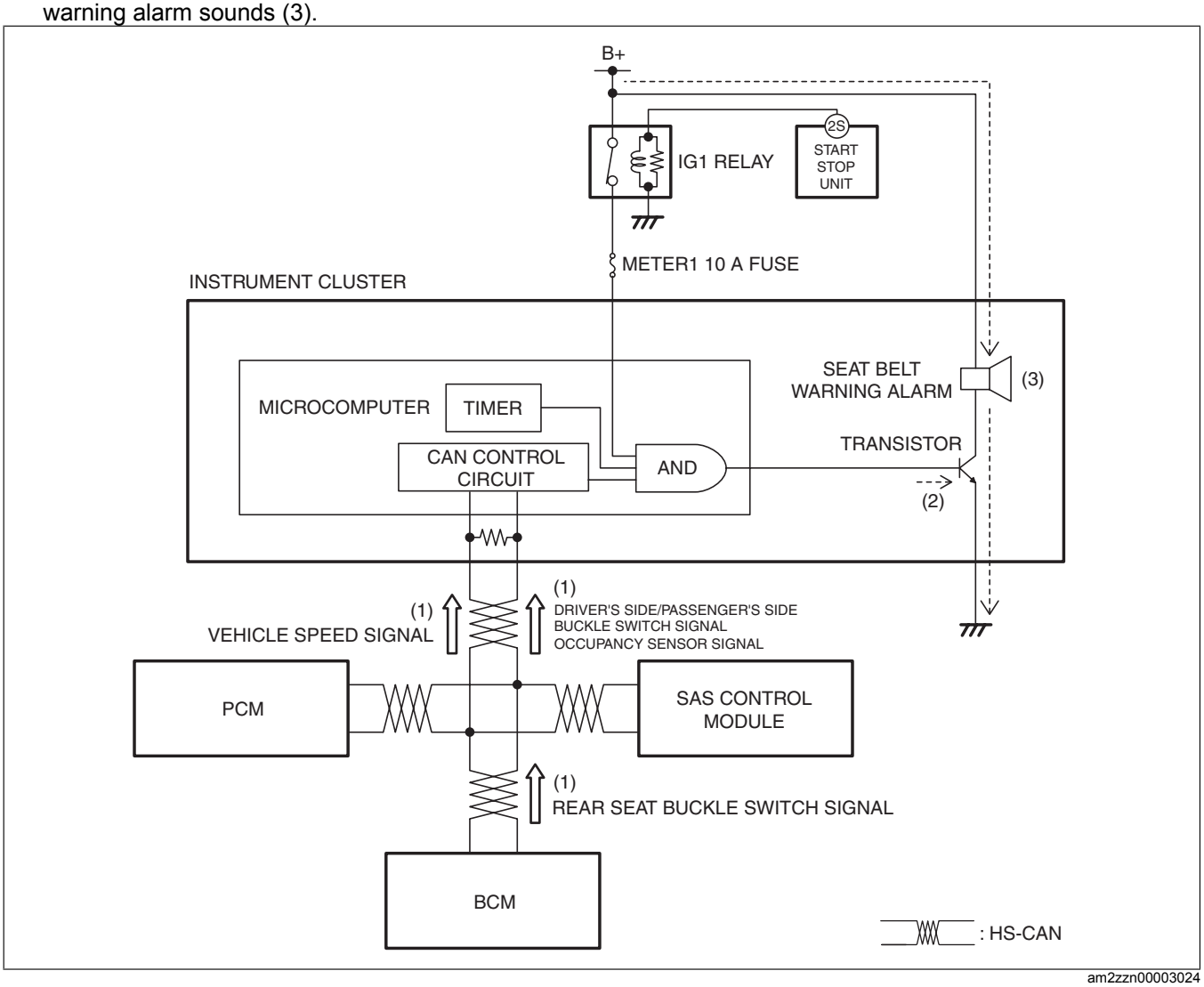
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Construction

- The seat belt warning alarm sounds using the buzzer built into the instrument cluster.

Operation

1. When the ignition is switched ON (engine off or on), the instrument cluster receives (1) a vehicle speed signal from the PCM, a driver's side/passenger's side buckle switch signal, occupancy sensor signal from the SAS control module, and rear seat buckle switch signal from the BCM via CAN communication.
2. The instrument cluster turns the transistor on (2) based on each signal.
3. When the transistor turns on, a ground circuit with the seat belt warning alarm is established and the seat belt warning alarm sounds (3).



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Fail-safe

- Not applicable